

Frequency distribution tables

By size and month

Task 1: Calculate the confidence intervals for men shoes sales in the USA, this time based on a bigger sample - 2015-2016.

Task 2: Compare the results with the confidence intervals when the sample was based just on 2016.

US	United States, 2015												United States, 2016												Mean
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5	6	7	8	9	10	11	12	15-'16
6	0	0	0	0	3	1	1	3	5	4	0	0	4	1	3	1	3	3	3	4	3	7	3	0	2.17
6.5	4	1	0	1	0	0	2	3	0	1	3	3	3	2	0	1	0	0	1	7	2	1	2	1	1.58
7	0	0	0	0	1	2	1	0	1	2	2	3	0	0	1	0	6	4	4	2	3	0	0	0	1.33
7.5	0	1	2	1	0	0	3	2	3	2	2	2	3	2	3	1	7	0	7	3	4	6	1	1	2.33
8	5	3	1	0	6	6	4	0	5	6	3	3	7	9	7	3	12	2	9	4	7	5	2	6	4.79
8.5	4	5	3	1	2	11	6	6	4	12	4	2	12	12	8	8	15	9	17	17	6	9	10	6	7.88
9	10	14	7	9	17	16	20	21	13	17	10	12	17	13	13	11	21	22	25	30	26	25	13	10	16.33
9.5	18	16	23	15	16	16	19	26	25	17	35	24	19	25	27	24	26	33	25	47	31	44	37	26	25.58
10	8	13	13	15	7	20	22	18	22	15	11	14	17	26	26	19	16	31	25	24	23	31	15	20	18.79
10.5	7	13	8	8	15	10	16	8	16	16	17	14	13	16	22	14	28	19	18	15	19	21	16	10	14.96
11	5	5	4	7	5	5	4	6	3	5	5	10	5	16	13	10	10	11	15	8	9	7	6	7	7.54
11.5	3	3	2	3	6	9	6	6	3	8	3	7	4	3	6	3	3	5	6	4	5	12	13	5	5.33
12	1	0	0	3	1	4	1	2	4	3	6	3	3	0	0	4	4	4	3	12	4	9	2	1	3.08
13	0	0	1	0	1	3	1	0	0	3	0	1	1	1	2	0	3	2	1	0	0	4	3	2	1.21
14	1	2	1	2	1	0	3	3	0	1	2	2	2	6	3	3	5	3	2	1	0	1	2	1	1.96
15	0	0	0	0	1	0	0	1	1	0	1	1	0	0	0	1	1	0	4	0	0	0	0	2	0.54
16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.00
Total	66	76	65	65	82	103	109	105	105	112	104	101	110	132	134	103	160	148	165	178	142	182	125	98	

Formula

$$SE = \frac{\sigma}{\sqrt{n}}$$

SE = standard error of the sample

σ = sample standard deviation

n = number of samples

$$ME = \text{reliability factor} * \frac{\text{standard deviation}}{\sqrt{\text{sample size}}}$$


The diagram shows two green arrows originating from the 'reliability factor' term in the ME formula. One arrow points to the z-distribution formula, and the other points to the t-distribution formula.

$$z_{\alpha/2} * \frac{\sigma}{\sqrt{n}}$$
$$t_{v, \alpha/2} * \frac{s}{\sqrt{n}}$$

Task 1

Standard error	Margin of error	95% CI		95% CI	
15-'16	15-'16	15-'16		2016	
0.39	0.81	1.35	2.98	1.78	4.05
0.34	0.70	0.88	2.29	0.45	2.89
0.34	0.70	0.63	2.03	0.33	3.00
0.41	0.85	1.48	3.18	1.64	4.70
0.60	1.24	3.55	6.03	4.14	8.03
0.94	1.95	5.92	9.83	8.28	13.22
1.26	2.61	13.72	18.94	14.50	23.17
1.77	3.65	21.93	29.24	24.95	35.72
1.33	2.74	16.05	21.53	19.30	26.20
1.02	2.11	12.85	17.07	14.57	20.59
0.72	1.49	6.05	9.03	7.53	11.97
0.58	1.21	4.13	6.54	3.63	7.87
0.58	1.20	1.88	4.28	1.60	6.06
0.26	0.53	0.68	1.74	0.75	2.42
0.30	0.62	1.34	2.57	1.32	3.52
0.19	0.39	0.15	0.94	-0.12	1.45
0.00	0.00	0.00	0.00	0.00	0.00

Task 2

Number of pairs, math round	Number of pairs, round up	Number of pairs (lesson)
3	3	4
2	3	3
2	3	3
3	4	5
6	7	8
10	10	13
19	19	23
29	30	36
22	22	26
17	18	21
9	10	12
7	7	8
4	5	6
2	2	2
3	3	4
1	1	1
0	0	0

	15-'16
n	24
t _{95%,df}	2.069

The number of sales of shoes have a tighter interval for 15 and 16 than for 16 only
16 may have been a good year for sales

t 0.025 Lookup>>